Hugo Marquerie 04/11/2025

Convergencia puntual dominada implica convergencia en $\mathcal{L}^p$

Proposición 1. *Sean* $(f_n)_{n\in\mathbb{N}}$ *funciones medibles* $f_n\colon X \to \mathbb{K}$ *en* (X,Σ,μ) *dominadas por* $g \in \mathcal{L}^p(X)$ *tales que* $f_n(x) \xrightarrow[n\to\infty]{} f(x)$ *c.t.p.* $x \in X$

$$\implies f_n \xrightarrow[n\to\infty]{\mathcal{L}^p} f.$$

***Demostración*:** Tenemos que $\forall n \in \mathbb{N} : |f_n(x)| \leqslant g(x)$ c.t.p. $x \in X$. Por tanto,

$$\forall n \in \mathbb{N} : |f_n(x) - f(x)|^p \leqslant (|f_n(x)| + |f(x)|)^p \leqslant (2g(x))^p \quad \text{c.t.p. } x \in X.$$

Luego, por el Teorema de la Convergencia Dominada, tenemos que

$$\lim_{n\to\infty} \|f_n - f\|_p^p = \lim_{n\to\infty} \int_X |f_n(x) - f(x)|^p \,\mathrm{d}\mu(x) \overset{\substack{\text{TCD}\\ \downarrow}}{=} \int_X \lim_{n\to\infty} |f_n(x) - f(x)|^p \,\mathrm{d}\mu(x).$$

Como $\lim_n |f_n(x) - f(x)| = 0$ c.t.p. $x \in X$, se tiene que $\lim_n \|f_n - f\|_p^p = 0$. ■

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